Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 — Unit-1 Semiconductor Diode · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_01 · Unit-1 Semiconductor Diode

1 UE26EC141A · Unit_01: Unit-1 Semiconductor Diode

This unit master organizes source material for Electronic Principles and Devices, Unit_01, officially titled Unit-1 Semiconductor Diode. It includes 11 video transcripts and 13 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26EC141A · Unit_01: Unit-1 Semiconductor Diode

Official catalogue entry: UE26EC141A, Unit_01 (Unit-1 Semiconductor Diode). Bucket folder present: yes. Videos: 11; slide decks: 13.

EV0000001

2 Introduction to semiconductors, intrinsic and extrinsic semiconductors

Study focus: Introduction to semiconductors, intrinsic and extrinsic semiconductors. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we are going to learn about unit one.

Key details

- In that, topic one, an introduction to semiconductor.
- The three semiconductors which is most commonly or most frequently used in the construction of electronic devices are germanium, silicon, and gallium arsenide.

- And we also see that the atomic structure of silicon and germanium.
- First, if you look at the atomic structure of silicon, the atomic number is 14, and it's represented in terms of SPDF model one S two, two S two, two P six, three S two.
- And if you start arranging with respect to the shell around, we call it as orbits.
- So the first orbit can have two, second can have eight, and the remaining four, that is three S two, three P two are at the outside.
- We call it as a valence shell, which has four valence electron.
- That means to say, they are not stable.
- But before that, they try to balance by themselves by moving from lower energy level to the higher energy level.
- There is a no gap at all.
- There's an overlapping in the case of a conductor between the conduction band and the valence band.
- Whereas in the case of a semiconductor, as we see, there is a little gap compared to the insulator and conductor.
- The term intrinsic is applied to any semiconductor material that carefully refine to reduce the number of impurities to a very low level.
- That is, we can call it as a pure form, made available through modern technology.
- Whereas extrinsic semiconductor are those that can be subjected to doping process with the impurities being added to that.
- Even in the upper part, even in the lower, even in the lower and also the side we see that there is a covalent bond where valence electrons are shared.
- With the increase in the voltage, the semiconductor act like a conductor where covalent bond is ruptured, with the temperature reduced act like an insulator where the covalent bonds are not ruptured.
- We know the four valence electrons are shared with the neighboring silicon atom, and one is free, the fifth valence electron of the antimony is free.
- And if you apply the voltage to this bonding arrangement, and covalent bond is ruptured, one more electrons will get free up.
- That means the number of electrons are more compared to the holes here.

Example or application from the lecture

- Similarly, any semiconductor material, if you look at example again, germanium 32, the outermost valence electron will be four again.
- For example, if you look at the first diagram here, as marked here, where the impurity added is the antimony impurity has been added, where outermost five valence electrons are there.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

No covalent bond is formed, so already there exists one bond.

Key details

- With the addition of an impurity over there, results in one hole.
- With the further change, means you say with applied voltage, one more covalent bond is ruptured, and a free electron is released and a hole is introduced over there.
- The positive charge material is formed with the introduction of a trivalent impurity with the three valence electron such as boron, gallium, and indium materials.
- We can construct our first solid state electronic device, that is a semiconductor diode, by sandwiching P-type and N-type material together to form a depletion region between the two.
- It is a fabrication of one material with majority charge carrier of electron to one with majority carrier of holes.
- How do we do a biasing of a diode?
- Where we go for forward bias and reverse bias.
- In a forward bias, a material type connected to the same polarity terminal of a voltage source.
- Whereas the reverse bias, a material type connected to a opposite polarity terminal of the voltage source.
- As shown in the diagram now, a P-type and N-type material sandwich with the depletion region form.
- P terminal and a N region is connected to the negative terminal, then it's said to be the forward bias.
- And if you reverse this, it's called as a reverse bias.
- We see the different type of diodes in the diagram here.
- Various companies have come up with the various structures of diodes here as we see.
- It could be a two terminal or a three terminal also sometime.
- IC-based diodes, which could be designed by the different companies.

Example or application from the lecture

- Next, let us see what is a diode, where diode is one which has a two airports.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

3 Semiconductor diode under no bias, forward bias and reverse bias conditions

Study focus: Semiconductor diode under no bias, forward bias and reverse bias conditions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000010 · EV0000011

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we will go with the topic three, Introduction to Semiconductor.

Key details

- In the last class we have looked at the diode and biasing of the diode, where we know that diode has two regions, P and N type, where P side is connected with the biasing.
- P side is connected to the positive of the terminal and N side is connected to negative of the battery terminal.
- P and N region not connected to the voltage supply.
- We call it as a no bias.
- That means to say, with no voltage applied to the diode, we see that the boundary, there is a depletion region with the charges being accumulated.
- This happens because at the instant the two metals are joined, the electrons and the holes in the region of the junction will combine, resulting in lack of free charges in the region near the junction, as shown, which we call it as a depletion region.
- And note in the figure that only particle displayed in this region are positive and negative ion remaining once the free charges have been observed.
- That means to say, because of the concentration gradient, the charge carriers will start move from.
- P side to N side and N side to P side.
- As the moment they start moving, they forms a charges which is been settled at the boundaries of the P and N, where negative charge is settled at the P region boundary and positive charge is settled at the N region boundary.
- This region of uncovered positive and negative ion is called depletion region.
- This is due to the depletion of free charge carrier in the region.
- So which we call it as a no bias condition.
- As the name indicates, a forward bias or 'ON' condition.
- The current that exists under the forward bias is called the forward bias current and is represented by ID in the diagram.
- As the name indicates, forward bias, we connect P region to positive terminal of the battery and N region to negative terminal of the battery.
- As the moment we connect it, we call it as an 'ON' condition, that is established by applying a positive potential to the P terminal and a negative potential to the N terminal, as shown in the diagram.
- The application of the forward bias potential that is VD we call here, will pressurize or will pressure the electron in the N-type material and hole in the.
- P-type material to recombine with ion near the boundary and reduces the width of the depletion region, as shown here.
- If compared to the previous diagram, we have a very big depletion region which is shown here.
- And because of the application of the voltage, the depletion region is reduced and we see only a few charge carrier.
- This happens due to the application of the voltage on both P and N region, and which results in the ID current which is a sum of the eye majority and the leakage current.

- This can be represented by a V-A characteristic, as shown, the forward bias current is shown here.
- There is a very less resistance is there in the forward bias and we see that there is a current flow with application of the voltage.
- This is about the forward bias.
- Then what is reverse bias?
- The reverse bias condition, the P side is connected to the negative terminal of the battery, N side is connected to the positive terminal of the battery with applied voltage VD.
- It's increasing due to the large number of free electron drawn from the positive terminal of the applied voltage.
- For a similar reason, the number of uncovered ion will increase in the P-type material.
- The net effect therefore is widening of a depletion region, as shown.
- This widening of the depletion region will establish too a great.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 05:00–09:59

a barrier for majority carrier to overcome, effectively reducing the majority carrier flow to zero.

Key details

- So the current that exists under a reverse bias condition is called the reverse saturation current and is represented as IS.
- And this also represented in the V-A characteristic as shown, with the initial applied voltage because of the depletion region will not be able to see the current flow which has been shown here.
- And once after the reverse breakdown region, once after the voltage called VBR, there is going to be a breakdown with the current increase as shown.
- And this could be realized from a reverse bias condition.
- So today's class we have seen with a no bias condition, forward bias condition and a reverse bias condition.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

4 Diode characteristics, Shockley’s equation

Study focus: Diode characteristics, Shockley’s equation. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Professor Ananda, Department of Electronics and Communication Engineering.

Key details

- Today, we are going to see the semiconductor diode characteristics and parameters and with some numericals.
- As we look at the forward bias here, initially for the applied voltage, the diode was off, and at particular voltage of Vf, the current start increasing.
- Then, we also come to another parameter called forward current rating.
- The forward current rating, which is I.
- It's the maximum current at which the diode starts conducting.
- It depends on the voltage at which the diode is on.
- So the current can vary based on the device, because it depends on the fabrication of the device.
- Then the maximum power expression P is equal to V into I.
- The first is breakdown voltage.
- Until there is a breakdown, there is a leakage current flowing, which is a reverse saturation current, which is less than the reverse saturation current, which is called as a leakage current.
- We need to remember this.
- Then, the reverse saturation current is this region, where the diode start conducting even with the reverse bias.
- That happens at a reverse bias voltage.
- It could vary based on the device.
- It could be five volts reverse saturation current.
- That depends on the fabrication again.
- Then we also need to calculate what the peak inverse voltage, the voltage at which the device starts conducting, which is a peak inverse voltage rating.
- And the diode representation symbol is this, where it looks like an arrow lying here.
- The arrow indicate the conventional current flow from P region to N region.
- Next, we also learn here the diode current expression or the Shockley's current equation.
- When Vd is positive, we represent Id is equal to minus Is.
- When Vd is negative, we represent Id is equal to minus Is.
- When Vd is positive, we represent Id is approximately equal to Is into E to the power of qVd by NmkBT, where Id and Vd are the diode current and voltage respectively, where q is the charge of the electron, where eta is the ideality factor, where eta N is equal to one for indirect semiconductor like silicon, germanium, et cetera.
- For a direct semiconductor, N is equal to two, where gallium arsenide, if you use gallium arsenide and indium phosphate.
- Where k is the Boltzmann constant, T is the temperature in Kelvin.
- We are going to use these k and the q values.
- Where the kT by q is also called as a thermal voltage, where kT by q is equal to 25.9 millivolts.

- At a temperature 27 degrees centigrade, a common temperature for a component in an enclosed operating system, determine the thermal voltage.
- As we know the thermal voltage, how to calculate?
- VT is equal to kT by q, where we need to calculate the T here.

Example or application from the lecture

- The Shockley's equation is also called as a diode equation, help us to determine the current at the given temperature, knowing the diode voltage and saturation current.
- That is given by Id is equal to Is into qVd by NqT minus one.
- Then, k is the Boltzmann constant given by value.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then substitute that here, where we know the k value, 1.38 into 10 to the power of -23, multiply with 300 Kelvin here, divide by 1.6 into 10 to the power of -19 coulombs.

Key details

- Then coming to the numerical two, calculate the thermal voltage when the temperature is 25 degree.
- The previous numerical was at 27 degree, now calculate at 25 degree.
- Similarly, coming to the next numerical.
- Calculate the forward bias current of silicon diode when forward voltage is 0.4 applied and the reverse saturation current is 1.17 and the thermal voltage is 25.2 millivolt.
- We need to calculate Id here.
- Since in this equation, ideality factor is not mentioned, it can be taken as one.
- Coming to numerical three.
- Vd NK T minus one.
- We have to find VT thermal voltage and reverse saturation current.
- Then substitute all the values for Id, for Is.
- For Id, you have to calculate.
- For Is, we need to calculate.
- The summary is the current should be less compared to the forward current.
- The reverse current is pico in 10 to the power of minus 12.
- Coming to the numerical four, calculate the applied voltage Vd if the diode current is six milliamps, thermal voltage is 26 millivolts, ideality is one, reverse saturation current is one.
- This numerical is not similar to the previous what we solved.
- Next, coming to the numerical five.
- Find the change in the voltage if current changes from 0.1 milliamps to 10 milliamps.
- We need to calculate 26 milli.
- We can directly use that value 26 ln 10, and the change, the I2 by I1.
- 10 milli by 0.1, so we get 0.143 volts.

- And finally, we can see the comparison of the diode characteristics of germanium, silicon, and gallium arsenide with respect to the forward cutting voltages.
- And also the reverse saturation current also has been shown.
- The breakdown voltage will vary for type of the device.

Example or application from the lecture

- Vd is equal to, which is kT by q, with a given value, they have already given 25.2, we know that n is equal to one, Vd is equal to 0.4.
- Substitute all the values here and calculate Id, which is given by 9.156 milliamps, and which is going to be solved in a class by the faculty.
- Given a diode current of eight milli and N is equal to one, find the reverse saturation current if the diode voltage is 0.5 and the room temperature is 25 degree centigrade.

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EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 10:00–14:59

We see that the breakdown voltage for germanium is very high, 100 volts.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

5 Zener and Avalanche breakdown, Temperature effects

Study focus: Zener and Avalanche breakdown, Temperature effects. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000030 · EV0000031

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we're going to learn about avalanche breakdown and Zener breakdown.

Key details

- Before that, we'll go for refreshing of the last topic, comparison of the diode characteristics of germanium, silicon, and gallium arsenide.
- We see that there are different voltages at which the different semiconductor device has different conducting voltages and with respect to forward bias and reverse bias.
- In the forward bias, we see the germanium, silicon, and gallium arsenide, which conduct at different voltages.
- The reverse bias also, the breakdown also occurs at different voltages.
- The first is avalanche breakdown for a normal diode we look at.

- The forward bias region is this, the reverse bias region is this.
- So in the reverse bias condition, as the voltage increases, the free charge carriers obtain velocity and associated kinetic energy and release additional carriers through collision with the other atomic structure.
- Hence, the covalent bonds are broken and electron-hole pair are generated.
- This electron will hit to the walls of an atom, as we see here, and acquire a energy from applied voltage.
- This will hit based on the energy acquired from applied voltage and produce more and more free electrons.
- This free electron again will hit the another atoms, and this cumulative process is called avalanche multiplication or ionization process.
- The Zener breakdown occurs when both P and N-type materials are heavily doped in a semiconductor diode, hence depletion region is narrowed down, as we see from the diagram here.
- When a small reverse voltage is applied across the diode, a very strong electric field is produced.
- Due to this electric field, covalent bonds breaks and produce a large number of free charge carriers.
- So coming to the comparison between the Zener breakdown and avalanche breakdown, first we start with Zener, and on the right-hand side we have avalanche.
- The first point is, the process in which the electron move across a barrier from valence band of the p-type material to the conduction band of n-type material is known as breakdown voltage, whereas the process of applying a high voltage and increasing the free electron.
- Coming to the second point, this is observed in a.
- Zener diode having a Zener breakdown, which varies from five to eight volts, whereas in the case of an avalanche breakdown, this is observed in the normal diode having a voltage greater than the eight volts.
- Number three, the valence electrons are pulled out into a conduction due to the high electric field in the narrow depletion region.
- Here, the valence electrons are pushed to the conduction due to the energy imparted with the accelerated electron which gain their velocity due to the collision with other atom.
- Coming to the fourth point, the increase in temperature decreases with the breakdown voltage.
- The increase in temperature will increase the breakdown voltage here.
- The VI characteristics of Zener diode breakdown has sharp value.
- The VI characteristics of avalanche breakdown is not sharp as.
- Zener breakdown, as we see from the diagram.
- Then in the Zener, it occurs in diode that are lightly doped.
- The effect of variation in the temperature across the semiconductor diode is observed in both forward as well as the reverse characteristics.
- The rise in temperature generates more electron-hole pair, thus conductivity increases and thus increase in the current.
- The PN junction diode parameter like reverse saturation current, bias current, reverse breakdown voltage, and barrier voltage are dependent on the temperature.

- Increase in temperature increases the carrier concentration.
- As a result, the knee voltage decreases, breakdown voltage increases, while the reverse saturation current also increases.
- This we can understand with the help of a forward bias and a reverse.

Example or application from the lecture

- Let us also consider the effect of temperature on VI characteristics of the diode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 05:00–09:59

Under reverse bias, the change in the temperature across the diode, the reverse saturation current of the diode increases with increase in temperature.

Key details

- That is, there is a rise of 7% for every degree rise in temperature for both germanium and silicon, and approximately doubles for every 10 degree rise in the temperature.
- The reverse breakdown voltage, VR, also increases with the temperature increase.
- Starting from right to the left, initially we have -75 degree centigrade, where the cut-in voltage is at this point.
- It may be around 0.1 or even less than 0.1.
- Whereas at 125 degree, the breakdown voltage is increasing, and reverse saturation current also doubles for every 10 degree rise in the temperature.
- Determine the value if the temperature is increased by 40.
- We know that for every 10 degree rise in the temperature, the reverse saturation current double.
- Determine the values at 40 degree centigrade.
- The difference is 15 degree.
- At 40 degree, what is the current?
- We need to calculate the current in the forward bias condition if the forward bias is 0.2 at room temperature.
- If the temperature is increased to 30 degree centigrade.
- Therefore, VT is 0.8 and actual current calculation is 1.83 amps.
- Find the current at 40 degree centigrade.
- We use the whole formula last time what we used.
- Substitute that and get the ID which is 281.49 milliamps.
- Next, fourth, the reverse saturation current of germanium is two pico at 27 degree.
- So we use the same principle concept where IS calculated first.
- Substitute the value and calculate the reverse saturation current.
- Determine the forward bias also.
- The forward bias voltage VA.

Example or application from the lecture

- In the example, the reverse saturation current of silicon diode is at 20 degree is 0.1 microamp.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

6 Ideal versus Practical diode, Diode resistances

Study focus: Ideal versus Practical diode, Diode resistances. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Professor, I'm in the Department of Electronics and Communication Engineering.

Key details

- Today, we're going to learn the ideal and practical diode characteristics along with the diode resistances.
- As we see from the diagram, we're going to learn about the equivalent representation of the diode, this has been written based on the characteristics curve.
- There are two types of practical diodes out there.
- There are two types of diodes representations are there.
- One is ideal diode.
- When an ideal diode is forward biased, as shown in the diagram here, it offers no resistance and act like a closed switch.
- Likewise, the ideal diode under reverse bias offers infinite resistance, hence it is a open switch.
- Coming to the practical diode as shown, the diode which is said to be forward biased, it starts conducting at a new voltage, which is 0.7 for silicon and reverse bias, no current, and this practical diode is considered as, again, the open switch, similar to that of an ideal diode.
- For germanium, it would be 0.3.
- So the next point, what we are going to understand is diode resistances.
- An actual diode offers a very small resistance, non-zero, when forward biased, and is called the forward resistance.
- Whereas it offers very high resistance, not infinite, when reverse biased, and is also called as a reverse resistance.
- Type of applied voltage or a signal will define the following resistance levels.
- Coming to the first DC or a static resistance, the application of an DC voltage to a circuit containing a semiconductor diode will result in operating point on a semiconductor diode characteristics curve, which is x-axis is VD, y-axis is.
- Typically, the DC resistance of a diode is for forward bias, 10 ohm to 80 ohm.

- Reverse bias, 10 megaohm.
- Coming to the AC or a dynamic resistances, the application of a small AC voltage to a circuit containing a semiconductor diode offers AC resistance.
- A straight line drawn tangent to the curve through Q point will define a particular change in voltage and a current.
- That change in the voltage and change in the current determine the AC or a dynamic resistance, which is shown that is RD is equal to delta VD by ID for this region of the diode characteristics.
- Coming to AC or a dynamic resistance, dynamic resistance can be found simply by substituting the quotient value of the diode current into the equation as RD is equal to 26 millivolts by ID, or RD dash is equal to 26 millivolts by ID plus.
- This RB is the bulk current body resistances.
- Typically, the AC resistance of diode in active region will range from 1 ohm to 100 ohm.
- Further, the average AC resistance can also be calculated with a diagrammatic from the diode characteristics as shown.
- If the applied input voltage is sufficiently large to produce a broad swing and a resistance determined by the straight line drawn between the two intersection established by a maximum and a minimum value of the input voltage is called the average AC resistance.
- That is, RAV is equal to delta VD by delta V at point to point.
- Then average AC, defined by the straight line between the limits of the operation.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

Determine the AC resistance ID is equal to 25 milli.

Key details

- Then compare the results of and to the DC resistance at the each level.
- 2 milli is the center point.
- Then substitute the value, we get 27.5.
- And substitute here, we get 2 milli.
- Coming to the point 3, compare the results of and to the.
- For a DC resistance at 2 milli, we have 0.7, we get 30, 50, RD is equal to 31.
- This is because in AC, we use a small value change, whereas in DC, we don't take a small value change, we take a large value change.
- Before we go for second numerical, there could be one derivation of expression for.
- AC or a dynamic resistance.
- We know that RD is equal to 26 millivolts by ID.
- How to derive this?
- KT ID plus IS.
- Then arrange, whereas we know that this is where IS reverse saturation, very less current.
- We know that K is equal to 11,600, TK is 298.
- After substitution, we get RD is equal to 26 millivolts by ID.

- This could be asked in the exam for some formulas.
- And the fifth numerical, here first they're asking, determine the DC resistance levels.
- In the previous numerical, we solved for AC, now for the DC at 2 milli, at 20 milli, and -10 milli, these are the reverse point.
- After substitution, we get the different values over there.
- In this comparison, for forward bias, the resistance is very less.
- Reverse bias, the resistance is very high, around 10 megaohms.

Example or application from the lecture

- Then coming to question number three, calculate the DC and AC resistance of the diode shown in the figure given below at a forward current of 10 milli, and also compare their magnitudes.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

7 Diode equivalent circuits Series diode configurations

Study focus: Diode equivalent circuits Series diode configurations. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today's topic will be diode equivalent circuit or diode approximation and series diode configuration.

Key details

- So coming to the diode approximation, is a mathematical method used to approximate the nonlinear behavior of the real diode to enable the calculations and circuit analysis.
- There are three different approximations used to analyze the diode circuits, namely the Piecewise linear equivalent circuits,.
- Simplified diode characteristics, and Ideal diode characteristics.
- Piecewise linear diode characteristics.
- Under forward bias, with respect to the V-I characteristics, the diode behaves as a practical voltage source with a magnitude equal to the knee voltage in series with the diode average series with the diode average AC resistance.
- This is shown the representation.
- Under reverse bias, the diode is an open switch, and no current flows here through it, act like an ideal diode reverse characteristics.
- The voltage drop across the diode is calculated by the formula VD is equal to Vb plus ID into RB.

- Coming to the Simplified diode characteristics, under forward bias, the diode turns on when a voltage VK or VB is applied across it.
- It turns off if the voltage applied is less than the VK.
- Thus, diode is represented by voltage of VK in a series with a closed path switch, as shown.
- Therefore, VD is equal to VB.
- Then coming to the reverse bias, it is an open switch similar to that of a practical reverse bias voltage.
- Coming to the Ideal diode characteristics here, we already know the diode is considered as a closed switch with a zero voltage drop.
- Then when in forward bias VD and open switch when it's in reverse bias.
- When the diode is connected in series, we call it as a diode configuration, a series diode configuration with a resistor.
- It is assumed that the forward characteristics or forward resistance of the diode is usually so small compared to the other series elements of the network that can be ignored.
- In general, the diode is in on state if the current established by the applied sources is such that its direction matches that of an arrow in the diode symbol and applied voltage across the diode is silicon VD greater than 0.7, germanium VD greater than 0.3, and gallium arsenide VD should be greater than 1.2 volts.
- And when it's reverse bias, diode is reverse connected to the positive voltage here and represented as an open circuit, and hence ID is current is zero here, and VR also is equal to zero volts.
- This is with respect to series diode configuration with a resistor.
- Assume the LED voltage as 0.7.
- We have a silicon and a LED here, and we have an output resistance 680 ohm.
- Then we need to apply the KVL here.
- We need to remember the KVL equation, which we learned from 12th standard.
- The sum of all the voltages across the closed loop network is always equal to zero.

Example or application from the lecture

- Let us see one by one today.
- Let us solve one numerical based on the series diode configuration.
- The question is, determine the Vout and ID for a given series diode circuit.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

V is equal to E minus VK1 minus VK2.

Key details

- We get 9.5 volts output.
- Then how to calculate ID?
- Since we know the voltage across this, so VR by R.
- That is Vout by R.

- Vout 9.5 divided by 680 ohm, which is 13.97 milliamps.

Example or application from the lecture

- So any numerical given in this fashion, if the diode is connected in series, we can calculate by applying a KVL first, calculate the output voltage, and take that output voltage divided by the total resistance, we get a current flowing through it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

8 Parallel and Series- Parallel configurations

Study focus: Parallel and Series- Parallel configurations. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today, we'll see parallel diode configuration.

Key details

- In the last class, we have solved the numerical and serial diode configuration.
- Today, completely, we look at the concept of parallel diode configuration with the numerical examples.
- The first question here is: Determine the Vout,.
- ID, ID1, and ID2 for the parallel diode configuration of the figure.
- I1 is this, ID2 is this, ID2 is this for the parallel diode.
- Since we have two diodes here connected in parallel, we call it the parallel diode configuration.
- Coming to the solution now.
- Then, how to calculate I1?
- The voltage applied between these two points, it is E minus.
- The applied voltage here and this voltage is 0.7.
- We get 10 minus 0.7, which is 9.3 divided by 0.33.
- Kelvin, which is 28.18.
- Coming to the ID1 and ID2, the potentials are same, and hence ID1 is equal to.
- Current also will be same, so I1 by I2, I1 by 2.
- So both current will be same now, because of the potential is same.
- 28.18 divided by 2, we get 14.09 milliamps.
- In case if the numerical has one series and one parallel, the previous one was both parallel.
- How to go about?
- Then we apply the KVL here.

- The diode is represented by 0.7 VD1, VD2.
- We are also able to apply the KVL this loop also.
- So first go with this.
- Then E applied voltage, leaving end is positive here.
- Then VD1, minus VD1 here, leaving end is minus V2, VD1, and leaving end is plus here, plus minus or the minus here.
- Then calculate the VD2 here.
- After substituting, we get 18.6 volts.
- Then calculate I2 across this.
- Since we know the voltage V2, calculate the value of the I2 with the resistance R2, we get 3.32.
- Then how to calculate this I1?
- The applied voltage here is 0.7.
- The 0.7 by 3.2 kiloohm will give you 0.212 milliamps.
- One thing we need to notice, each numerical may not be same.
- Ohm's law, then we'll be able to get the answer.
- Coming to one more numerical, where diodes are connected in process.
- We have a red LED connected 1.8.
- I think this numerical we solved.
- Just to refresh to people, the diode can be connected in series.
- We already solved this in topic four or five.

Example or application from the lecture

- Given, if you measure the voltage from this point to this point, this is 0.7.
- Coming to the question number three or the second numerical example for series parallel diode.
- Then the total current given by ID2 plus I1 is equal to I2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 05:00–09:59

When applied voltage is positive here, so the diode is reverse bias now.

Key details

- Obviously, so we get V output is equal to 0 over there.
- Then how to calculate the voltage across VD2?
- VD2 is 12 volts now.
- How it is 12 volts?
- Apply the KVL to the loop.
- We get VD2 is equal to E minus VD1 minus VD2, which is the 12 volts.
- Where Vout is equal to 0, yes, we'll be able to get, and VD1, we have 0.7 here.
- For ideal diode, it is 0, but 0.7, we get here 11.3 volts.
- Sorry, it is 11.3 volts we need to get.

- And Vout is equal to 0.
- I have here, which would be asked for one mark.
- See, it's not that the question could be asked for only four marks.
- A question could be asked for one mark also.
- Determine the current for each of the configuration.
- For this configuration, calculate the I.
- I is equal to, apply the.
- KVL, or first look at the negative voltage is connected here.
- When you apply negative, diode is reverse bias.
- When it's reverse bias, the current I is equal to 0.
- Since the diodes are connected in reverse here, the center branch will open here.
- The current will pass only through the resistor.
- Then coming to the few more numerical on parallel diode configuration.
- If the diodes are connected, one in same configuration, other in different configuration, one diode is ON, other diode is OFF.

Example or application from the lecture

- In the given question, they are asking us to calculate the VD2.
- C, calculate the current in the given configuration.
- Try to follow the slides and solve the numerical as per the given question, and we also will solve in the class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

9 Logical Operations using diode

Study focus: Logical Operations using diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000062 · EV0000063

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Dr.

Key details

- Anand, Department of Electronics and Communication.
- Today, we're going to see the logical operations.
- A logical operation is an operation that acts on a binary number to produce the result according to the laws of Boolean logic.
- A logic gate is a basic building block of digital circuits.

- Logic gate is considered as a device that has the ability to produce one output level with a combination of input levels.
- The function is represented f and Y is the result.
- In continuation, logic gates are implemented by using diode transistors and by other several devices.
- This logic gate can also be considered as electronic circuits.
- The logic gate opened by diode is called diode resistive logic.
- Inputs and outputs of logic gates are in two levels, which are termed as high and low, or true or false, or ON and OFF, simply one and zero.
- Coming to the next step.
- For all logic gates, a table with all the combination is listed below.
- The combination of input variable and the corresponding output is termed as truth table, which is shown here.
- It explains how the logic circuit output responds to various combination of logic level at the input.
- AND operation is explained by the following truth table, where.
- AND operation is A.B.
- When inputs A and B are zero, zero, B results in zero.
- Input zero and one multiplied to get zero.
- One and zero, zero.
- One and one is one.
- This is also explained by a Venn diagram here.
- The common is being highlighted here with the red color.
- OR gate produces an output of logic one as shown in the diagram.
- Even if one of the input is one state and produces an output of zero state if all the inputs are in the logic zero state.
- OR operation is explained with the following table.
- With zero, zero, we get zero.
- With zero and one, and one and zero, and one and one, we get all one states.
- The two diodes are connected in parallel here, and they're connected in series with a resistance with 1K.
- With the input E is equal to 10, with input.
- E1 and E2 here.
- And that we can represent using a truth table with the input V1 in the one column,.
- V2 second column, and V output in the third column.
- The input V1 is also called as an input A, and similarly input two respectively with respect to V1 and V2.
- When both the inputs are zero, zero, diodes are not forward biased and hence Vout is equal to zero.
- If one of the input is 10 volts, so as shown in the diagram here, the first V input is 10 volts here, the second is zero, the diode one is forward biased hence we represent 0.7.
- And what the difference between the applied voltage here and we calculate the Vout.
- It's not 0.3, it's 0.7.

- Here diodes are reverse connected compared to the previous one, and we connect a voltage E is equal to 10 volts here.
- And applied voltage E1 and E2.
- Because the difference between here is 0.7, so we get 0.7.

Example or application from the lecture

- So we can build an example of AND, OR NOT functions.
- Now consider the first condition here, zero, zero.
- If one of the voltage we consider 10 volts here, so we consider this 10 volts and the second voltage is zero, diode is on.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 05:00–09:59

The previous case also diode was on, so now second diode is on, and hence the difference in the voltage is 0.7 we get.

Key details

- If both the diodes are applied with E1 and E2, 10 volts, then the both diodes are off, and hence the voltage across this is 10 volts.
- AND with the help of a diode.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

10 Diode Clippers : Series using Ideal diode

Study focus: Diode Clippers : Series using Ideal diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today, we're going to see the clippers.

Key details

- Clippers are those circuits which are made up of electronic components such as diodes, resistors, and DC bias voltages, which are used to remove a part of the waveform without disturbing the other part of the waveform.
- That means to say, in case you have a square waveform, as shown in the figure here.

- If I need only the positive part of the waveform and I don't want the negative part, using the clipper circuit, we can get only the positive part.
- Further, the series clipper can be grouped into unbiased series clipper, biased series clipper, and in parallel, it's been classified into unbiased parallel and biased parallel.
- The series configuration is defined as one where diode is in series with the load.
- Figure 1A, we are using a series unbiased clipper.
- First of all, what is this unbiased?
- Unbiased means there is no DC supply used in the circuit.
- Only we have in this block, we have only diode and the resistor.
- In case if there is any bias voltage here along with the resistor or along with the diode, other than the supply voltage.
- Other than the supply voltage, if we have a voltages here, a DC voltage, then it is biased.
- If the diode is connected in series with the resistor, it is called as a series configuration.
- The response of the series configuration of Figure 1 to the variety of alternating waveform is as shown in the figure B.
- That means to say, variety of the input is first, in the diagram here, which I'm pointing, it is a square waveform.
- The output of the series clipper will be only the positive off cycle, negative has been clipped off.
- Similarly, we have next, in the place of square, we have a triangle waveform.
- That means we need to understand that there's no boundary, where we can also use a sine waveform, the sawtooth waveform, et cetera, as an input signal and based on the circuit, like a clipper, the output variations can be obtained.
- Figure shows here.
- Figure shows a series clipper with a DC bias supply.
- This is what I was referring previously.
- When there is a DC bias here, it is called as a series clipper with a bias or a series clipper with a DC supply.
- This configuration is called as a biased series clipper.
- The addition of a DC supply to the network as shown in the figure can have a pronounced effect on an analysis of a series clipper configuration.
- That means there could be some change in the output of which the DC supply can be in the leg between the supply and the output or in a branch parallel to the output.
- It can be at any place.
- One, it can be between the input and the diode, or diode and the resistor, or resistor and the output.
- It can be anywhere.
- The output response is not as obvious because the DC can aid or work as a source voltage.
- That means the DC source, which is there here, can aid in the sense it can have an addition with the voltage applied here or it can subtract with the voltage applied here.
- That we will understand with the analysis in the next slide.
- The step one as shown here, take careful note of where the output voltage to be defined.

Example or application from the lecture

- Example in the.

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EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

So we need to first note that where we need to calculate the output voltage.

Key details

- Try to develop an overall sense of the response by simply noting that how the input supply affects the conventional current direction through the diode.
- This means to say, check on the input voltage, how it has been applied here, whether it's a positive or a negative.
- And since we are working with an ideal diode here.
- It is an ideal diode.
- Entire discussion will be with the ideal diode.
- Means to say, the diode is on, even if the application of the voltage is zero.
- If it's more than zero.
- Whereas for a non-ideal diodes, it will be 0.7.
- And here, one more thing we also we need to remember, Vi is greater than V.
- Otherwise, the diode will not be on.
- Obviously, this should be there for the diode to be on.
- That means to say, diode is on.
- How do you represent when the diode is on?
- By a close switch.
- That is, VD is equal to zero.
- For ideal diode characteristics, if I look at the ideal diode characteristics, this is what we looked at the previous class.
- This is V and forward bias.
- And diode will start conducting at the zero.
- That's what we have written here, VD is equal to zero.
- Then the current ID,.
- VD is equal to zero, diode is on, but how about the current?
- So note that the substitution of a short circuit equivalent for a diode is as shown in the figure three we discuss now, and the fact that the voltage across the resistor is zero, because the diode current is zero milliamps.
- The result is Vi minus V is equal to zero or Vi is equal to V and is applied a transition voltage.
- This permits drawing a line on the sinusoidal supply voltage as shown in the figure four.
- As shown in the figure four, on state.
- To define the region where the diode is on and off.
- Figure five, the output voltage is defined by, apply the KVL here.
- Apply the KVL in this direction.
- Vi, this is Vi plus Vi.

- Vout is equal to...
- What is Vout voltage?
- Vout is equal to Vi minus V.
- Then for the off-region.
- This is off-region and if you go back, here this is the off-region.
- Off region, diode is in open circuit and ID current is equal to zero milliamps and the output voltage is zero.
- Since there is no current there, output voltage is zero.
- This is shown in the figure.
- It may not be the previous diagram.
- It can also be written in this diagram with the equivalent scales.
- So for on condition, equation one can be used to find the output voltage with the applied voltage as its peak value.
- Output voltage VOUT is plotted as shown in the figure six.
- This is the figure six, how we plot it.

Example or application from the lecture

- Means to say, for example here, this DC voltage, what we apply, and along with the input voltage, whether it make the diode on or off.
- That is given as equation number one here.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

Now let us solve a numerical example one.

Key details

- The question is, determine the output waveform for a sinusoidal input of figure seven.
- This is figure seven.
- I have a sinusoidal signal here.
- VI 20 volts is a peak to peak of one cycle.
- Step two, the positive region of the VI and the DC are both working in the same direction to turn on.
- So the applied total voltage is 10 plus 5 because both are in the same direction.
- We have plus is there, and we have plus is there, it get added.
- The result is that we can safely assume the diode is in on state for entire region of positive cycle voltage.
- Once the supply goes negative, it will have exceeded the.
- DC supply voltage of 5, therefore it would turn off.
- What is the transition model?
- If VI and the DC voltage is greater than the diode cutting voltage, that is VD is equal to zero, then the diode is in on condition.
- Diode is on.

- Therefore, we apply the KVL over there.
- If you apply the KVL over there, VI plus 5 is equal to zero, so VI is equal to minus 5 volts we get.
- Minus 5 volts.
- For a voltage less than 5, the diode is in open circuit and the output is zero, as shown in the sketch of the VOUT.
- This is what we want to prove here.
- Using the figure number nine, we find that for a condition when a diode is on and the diode current is established, the output voltage will be following.
- So the analysis of the clipper circuit with the square wave input is actually easier than the sinusoidal because only two levels are to be considered.
- The transition is important.
- It is clipped off.
- For a transition from 0 to T/2, the figure 11.
- Result, the diode is in short circuit state, that is on.
- If we apply the KVL, we get VOUT is equal to 20 plus 20, 25 volts is the applied voltage.
- Then for VI is equal to -10 during the negative cycle, diode is off.
- That is VOUT is equal to zero.
- Unless we get a positive cycle only, and during the negative cycle, we won't see the output waveform.
- It is clipped off.

Example or application from the lecture

- Then coming to the circuit here, we have a biased clipper here, series biased clipper, where resistor is connected in series with the diode and voltage 5 volts is given, and VI is equal to 20.
- That means to say in the place of sine, if you have a square, only two places if you consider it is more than, whereas here we need to consider for all the places here.
- So this is the solution for example number one, where we see that during the positive cycle, diode is on, during the negative cycle it is off.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

11 Diode Clippers : Parallel using Ideal diode

Study focus: Diode Clippers : Parallel using Ideal diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000082 · EV0000083

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Professor Ananda, Department of Electronics and Communication Engineering.

Key details

- Today, we're going to see the parallel clipper, but the last class we have seen the series clipper.
- The series clipper are those which is used to clip off the waveform that is not required and without disturbing the other part of the waveform.
- The analysis of the parallel configuration is very similar to that of the series configuration.
- The other end of the resistor is connected to the diode.
- When the diode is on, we represent it by a short circuit.
- When the diode is off, represented as an open circuit.
- When it's represented as an open circuit, the voltage across the diode is same as the input applied.
- The same thing applies to any input in the place of a square if you use the triangle as shown.
- Determine the Vout for the network of the figure 16.
- DC voltage, V is equal to 4 volts, with the positive side connected to the positive side of the diode and negative side connected to the ground.
- And the diode positive side is connected to the resistor, where the voltage applied.
- VI is applied to the one end of the resistor, other end is connected to the supply voltage.
- DC voltage, and we see that the negative is in series with the positive.
- That means the applied voltage here will get minus with the V.
- That means to say...
- So we'll try to understand that with the next step.
- Then, in other words, when the diode is on, the output will be 4 volts.
- This we need to prove now.
- Then when the diode is in open circuit, that is the off state, the current through the series network will be 0, and the voltage drop across the resistor will be 0.
- That will be Vout is equal to VI.
- During the positive half cycle, transition from 0 to T/2, diode is on.
- How do we represent?
- 4 volts there.
- And the diode is 0, and transition 0 instead of transition.
- That is why VI greater than 4 volts, Vout=VI.
- That means to say, if VI is greater than 4 volts, then VA greater than 4 volts, diode is off.
- How do you represent that?

Example or application from the lecture

- Let us have a refresh of that.
- Now let us go ahead with the parallel clipper, first without bias, then with bias.
- So let us go with the clipper with a bias, with an example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 05:00–09:59

Greater than 4, less than 4, 4 volts, greater than 4.

Key details

- VK is equal to 0.7.
- During the transition voltage, using the ID is equal to 0 or VD is equal to 0.7 volts and draw current in the network as shown in the figure 19.
- In the previous case, it was 0, VD was 0.
- The total voltage applied is, we'll add or subtract, so 3.3.
- It's subtract here because I have minus plus or plus minus.
- It's 3.3.
- In the previous case, it was 4 volts there.
- Whereas now, the total voltage, sum of the voltage here is 4 minus 0.7 is 3.3 we get.
- Then during the negative half cycle what we notice is, same as an input.
- The input voltage is 16 here, and hence we follow up to 16 the same waveform, and T/2 to T, the same 3.3 will be continued.
- The summary of the series clipper and the parallel clipper.
- In a positive clipper, the positive part is missing here.
- The circuit is this.
- Hence we call it as a negative clipper.
- Then, coming to the bias.
- The positive was missing in the previous case.
- Now positive is there, but it got clipped off because of the voltage available here.
- Presence of the voltage over here.
- This is because of the transition happened because of the clipper.
- Because of the voltage applied there.
- Then come to the simple parallel clipper with ideal diodes.
- The positive part is missing, positive clipper.
- Negative part is missing, negative clipper.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

12 Clampers using Ideal diode

Study focus: Clampers using Ideal diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Dr.

Key details

- Associate Professor in the Department of Electronics and Communication.
- Today, we'll see clamper.
- Last class, we have seen the clipper circuits, the series, and parallel configurations.
- A clamper is a circuit with the diode resistors and capacitor that shift the waveform to a new DC level without altering its shape.
- It's also known as a DC restorer.
- The clamping circuits have capacitor connected directly from input, as shown in the figure here.
- Resistor element connected in parallel with the output signal.
- The diode is also in parallel with the output signal, but may or may not have a series DC supply as added element.
- Examine the response of portion of an input signal that will forward by the diode.
- As usual, like how we did for a clipper to the positive of cycle where the diode is on.
- During the negative cycle, where the diode is off, we need to find out.
- For figure two, we see that the applied voltage here positive, and the diode is on, and the capacitor also is charging from plus to minus.
- VD is equal to zero.
- The short circuit for a diode results in Vout is equal to zero for T is equal to zero to T/2.
- That is zero line.
- The zero line is there.
- Step two, during the on period of the diode, assume that the capacitor will charge up to instantaneously to a voltage level determined by the surrounding network.
- That's why we represent here V.
- Step three, assume the capacitor retains its voltage while the diode is off.
- That means to say with the negative cycle of the input, the negative cycle diode is off, and we are represent as open circuit here.
- We need to remember, capacitor is retaining the charge over there.
- Whereas during the positive cycle, capacitor is charged to voltage.
- It is charged.
- Then the step five, check the voltage swing of the output matches that of an input.
- How we get this minus 2?
- Apply the KVL, this loop actually.
- Vout is equal to minus 2V.
- Here represented as 2V.
- This is because of the capacitor.
- And hence, during the negative of cycle, the diode is off.
- Since it's retaining the voltage, when you apply the KVL, it add it to the applied voltage.
- And hence, Vout is equal to minus 2 volts.
- I have a series of the input here, not.

Example or application from the lecture

- Today, let us see on clipper.

- Let us see.
- Then let us solve a numerical for a different set of inputs being given over there.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

We have a two cycles here being given here and is represented as zero to.

Key details

- T1, T2, T3, and T4.
- Yeah, we'll go to this.
- Therefore, from T1 to.
- T2, there's a negative off cycle.
- Whereas from zero to T1, input voltage Vi make the diode off, where the diode is off open circuit and Vout is equal to Vi.
- But now from T1 to T2 of the wave, since diode is in short circuit state, the network appears as shown in the figure 1.2.
- So Vout is equal to five volts now, because here.
- VD is equal to zero.
- This will add to the 5 volts.
- This is for the period T1 to T2.
- If I apply the KVL across this, minus 20, the input voltage, plus.
- VC, minus 5 volts, 0.
- Twenty-five volts.
- Coming to step three.
- For the period from T2 to T3, the next half cycle.
- The network will appear as shown in the figure 1.3 now.
- That means that the diode is open circuit, and this part is, even though it's present, it is absent.
- But before that, we need to plot with a time duration.
- Then the resulting output appears in 1.4, the input signal here.
- This is how it look like the output for a clipper circuit.
- So for a short circuit, so for T1 to T2 first.
- We need to check that, because along with the 5 volts, we have 0.7 negative there.
- Then similarly, the charge stored, apply the KVL across the...
- Because with the capacitor the short voltage is 24.3.
- The previous case it was 25, and hence the voltage was 35 total.
- 0.7 is reduced.
- So now, so the voltage reduced value of the first is 4.3, and it is 34.3, and again 4.3 is for the cycle from T3 to T4.
- Then there are various clamping networks are there, and we have to recognize.

Example or application from the lecture

- Let us start.
- And there is also a given supply voltage here.
- Let us go back to the previous circuit.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

The positive is clipped off here.

Key details

- The negative part is been added here.
- The positive is there.
- The negative is added over there.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

13 Source slide deck: Experiential learning Unit 1

Study focus: Source slide deck: Experiential learning Unit 1. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000489 · EV0000490

SHORT-NOTE RECAP

Study notes · Source slide deck: Experiential learning Unit 1

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ELECTRONIC PRINCIPLES

- AND DEVICES (4-0-0-4-4)
- Department of Electronics and Communication

Slide 2 · ELECTRONIC PRINCIPLES AND DEVICES

- Physically show all components in class: Passive and active components with
- Signal-generator and measuring devices like Multimeter and Digital-Oscilloscope.
- Department of Electronics and Communication.

Slide 3 · ELECTRONIC PRINCIPLES AND DEVICES

- (i) (a) List of Passive Components

- 1. Resistors : 220Ω, 680Ω, 560Ω, 1MΩ
- 2. Capacitor : 1µF, 22µF

Slide 4 · (i) b List of Active components

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 5 · (i) c List of Measuring instruments & Functionality

- (1) Multimeter : To measure Voltage, Current, Resistance etc
- (2) Cathode ray oscilloscope(CRO): To measure voltage(p-p) and
- current and also to identify the type of waveform and compare it.

Slide 6 · Activity 1: Measure the value of Resistor,

- Capacitor and POT using Multimeter

Slide 7 · Activity 2: (i) Identify the P-side and N-side in Diode

- (ii) Identify the terminals of BJT
- (ii) Identify the terminals of BJT
- (i) Identify the P-side and N-side in Diode

Slide 8 · Activity 3:How to set the input the input value like

- voltage & frequency in signal generator using CRO

Slide 9 · Activity 4: Measure the voltage & current by

- forward biasing Diode/BJT using Multimeter

Slide 10 · Activity 5: Breadboard Usage

- A solderless prototyping board used to build temporary circuits without soldering.
- Power rails run along the sides and are used for VCC
- Power OFF while wiring • Keep leads short • Check connections before

Slide 11 · Department of Electronics and Communication

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000489 · EV0000490 · EV0000491 · +30 source units in pack.json

14 Source slide deck: Experiential learning Unit 1

Study focus: Source slide deck: Experiential learning Unit 1. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000522 · EV0000523

SHORT-NOTE RECAP

Study notes · Source slide deck: Experiential learning Unit 1

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ELECTRONIC PRINCIPLES

- AND DEVICES (4-0-0-4-4)
- Department of Electronics and Communication

Slide 2 · ELECTRONIC PRINCIPLES AND DEVICES

- Physically show all components in class: Passive and active components with
- Signal-generator and measuring devices like Multimeter and Digital-Oscilloscope.
- Department of Electronics and Communication.

Slide 3 · ELECTRONIC PRINCIPLES AND DEVICES

- (i) (a) List of Passive Components
- 1. Resistors : 220Ω, 680Ω, 560Ω, 1MΩ
- 2. Capacitor : 1µF, 22µF

Slide 4 · (i) b List of Active components

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 5 · (i) c List of Measuring instruments & Functionality

- (1) Multimeter : To measure Voltage, Current, Resistance etc
- (2) Cathode ray oscilloscope(CRO): To measure voltage(p-p) and
- current and also to identify the type of waveform and compare it.

Slide 6 · Activity 1: Measure the value of Resistor,

- Capacitor and POT using Multimeter

Slide 7 · Activity 2: (i) Identify the P-side and N-side in Diode

- (ii) Identify the terminals of BJT
- (ii) Identify the terminals of BJT
- (i) Identify the P-side and N-side in Diode

Slide 8 · Activity 3:How to set the input the input value like

- voltage & frequency in signal generator using CRO

Slide 9 · Activity 4: Measure the voltage & current by

- forward biasing Diode/BJT using Multimeter

Slide 10 · Activity 5: Breadboard Usage

- A solderless prototyping board used to build temporary circuits without soldering.
- Power rails run along the sides and are used for VCC
- Power OFF while wiring • Keep leads short • Check connections before

Slide 11 · Department of Electronics and Communication

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000522 · EV0000523 · EV0000524 · +30 source units in pack.json

15 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000555 · EV0000556

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 1 from 2025 [1]__10viKVTE9W.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 3/25/25, 10:58 AM ESA - Jan 2025 - UE24EC141A (set- 1) about:blank 1/5 (8.0 Marks) (9.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan-Feb UE24EC141A - Electronic Principles And Devices Total Marks : 100.0 1.a. (i) Determine the dynamic resistance of a diode at a temperature of 40 C and a forward bias voltage of 0.6V, if the reverse saturation current is 5pA at 300K. n = 1 (4M) (ii) Determine I and V in the circuit shown below (4M) 1.b. What are the different approximations used to represent the diode? For each of them draw the equivalent circuit and VI Characteristics o D 0 5.77 mA 0.46107 m ohms

Archive page 1 from 2025 [2]__1LrJQqKVte.pdf; unit match candidate score 0.5833; source text/OCR may be partial. 3/25/25, 10:58 AM ESA - Jan 2025 - UE24EC141A (set- 2) about:blank 1/7 (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan-Feb UE24EC141A - Electronic Principles And Devices Total Marks : 100.0 1.a. Describe the Temperature Effects on V-I Characteristics of a Diode with Forward and Reverse biased and suitable diagrams? (Diagram-2 M + Forward bised - 2 M + Reverse biased -2 M) 1.b. Determine the voltages and current through 20Ω and 10Ω resistor and also the total Current (I) for the configuration (2M + 1M +1M) 19.3 1.93 19.3 0.965 2.895

EV0000555 · EV0000556

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9968; only the JSON segmentation supplies the displayed timestamps.
Introduction to semiconductors, intrinsic and extrinsic semiconductors transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__183ac314557f22c2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9939; only the JSON segmentation supplies the displayed timestamps.
Semiconductor diode under no bias, forward bias and reverse bias conditions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__1070a788810cfac7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode characteristics, Shockley's equation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Diode characteristics, Shockley's equation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/003_Diode_characteristics_Shockley_s_equation__09b9cf730275f786.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zener and Avalanche breakdown, Temperature effects transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Zener and Avalanche breakdown, Temperature effects transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/004_Zener_and_Avalanche_breakdown_Temperature_effects__83cce28872db58b1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Ideal versus Practical diode, Diode resistances transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/005_Ideal_versus_Practical_diode_Diode_resistances__3b08a07358669379.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode equivalent circuits Series diode configurations transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9966; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Parallel and Series- Parallel configurations transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Parallel and Series- Parallel configurations transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/007_Parallel_and_Series-_Parallel_configurations__f6ab675a2f96218e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9977; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode Clippers : Series using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Parallel using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9977; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode Clippers : Parallel using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/010_Diode_Clippers_Parallel_using_Ideal_diode__b4558739e39a139a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Clampers using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/011_Clampers_using_Ideal_diode__a50c9b8c3f829f04.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Introduction to semiconductors intrinsic and extrinsic semiconductors	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Introduction to semiconductors intrinsic and extrinsic semiconductors	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Introduction to semiconductors intrinsic and extrinsic semiconductors (converted PDF; original hash 10836acb38251f1a6456c6ee7c31fb424695e79f4c7ec0bf52052c6c790e33e0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/001_Introduction_to_semiconductors_intrinsic_and_extrinsic_semiconductors__349c88a00d515771.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Semiconductor diode under no bias forward bias and reverse bias conditions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Semiconductor diode under no bias forward bias and reverse bias conditions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Semiconductor diode under no bias forward bias and reverse bias conditions (converted PDF; original hash 7316e46912b815d4fabf620c52e93e433e55f0cd560ff2059481868325749cf2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/002_Semiconductor_diode_under_no_bias_forward_bias_and_reverse_bias_conditions__63662d27d393ceed.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode characteristics Shockley s equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Diode characteristics Shockley s equation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode characteristics Shockley s equation (converted PDF; original hash 8175b4e63ef1b61c0f6e35bdc122fae79376c17503d529a988b2efaf83987ea2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/003_Diode_characteristics_Shockley_s_equation__87b47de24c62e1d8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Zener and Avalanche breakdown Temperature effects	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Zener and Avalanche breakdown Temperature effects	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Zener and Avalanche breakdown Temperature effects (converted PDF; original hash cdec8f804bb21361720719598181 4ec3927fd064950ca7549ea330aa7 57ea27d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/004_Zener_and_Avalanche_breakdown_Temperature_effects__e2302d9472a3245c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Ideal versus Practical diode Diode resistances	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Ideal versus Practical diode Diode resistances	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Ideal versus Practical diode Diode resistances (converted PDF; original hash fd04b9a2689d48164c1b684fadfa9dddd18ded5098eac76a684a6ac33d1c120d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/005_Ideal_versus_Practical_diode_Diode_resistances__14c863657e4b9e42.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode equivalent circuits Series diode configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Diode equivalent circuits Series diode configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode equivalent circuits Series diode configurations (converted PDF; original hash 51f1afe81317c1c49f809f003fd54686c0951fcd952ea07a58d6eb4d0b9c5385)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/006_Diode_equivalent_circuits_Series_diode_configurations__5453dc2f4e4957f0.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Parallel and Series Parallel configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Parallel and Series Parallel configurations	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Parallel and Series Parallel configurations (converted PDF; original hash 27d54c28dbffe0fd12cd5a4ffa3f28944f3957d04c83a2412a72e88ca1ef3835)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/007_Parallel_and_Series-_Parallel_configurations__aa18f7794f270520.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Logical Operations using diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Logical Operations using diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Logical Operations using diode (converted PDF; original hash 4201923b33b3c6c2e2b1586fd2dae0eef75e584057a8c7aadd60c41f1669e4da)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/008_Logical_Operations_using_diode__69541ca2fb5e21d2.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode Clippers Series using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Diode Clippers Series using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode Clippers Series using Ideal diode (converted PDF; original hash 634c9aee0eedd6dd35f4d1eb18596f004701f9fea9f4f840f40aa146edbe3697)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/009_Diode_Clippers_Series_using_Ideal_diode__7fc19e7302be29a7.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Diode Clippers Parallel using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Diode Clippers Parallel using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Diode Clippers Parallel using Ideal diode (converted PDF; original hash c03093630fb1f356cc8af8dae8671a4d77ba2c903c269e9d6be09f45f6b9fbe8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/010_Diode_Clippers_Parallel_using_Ideal_diode__20dfefa028bcfd77.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Clampers using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Clampers using Ideal diode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Clampers using Ideal diode (converted PDF; original hash 290846c250a20e98abeedf97381f3bac5d5a53b1a75d36fde81637f2eda3c9ca)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/011_Clampers_using_Ideal_diode__f7d3c8caf08bbb8f.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Unit 1 (converted PDF; original hash 59479989b0f50a3c5fa2b0df49cc737f7c6d0cf73fd5cbe45bb06dbebf4fc5d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/013_Experiential_learning_Unit-1__3854d81f81e55ac8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Experiential learning Unit 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Unit 1 (converted PDF; original hash 59479989b0f50a3c5fa2b0df49cc737f7c6d0cf73fd5cbe45bb06dbebf4fc5d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/Slides/014_Experiential_learning_Unit-1__0d18720b1b592cfc.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: 2025 [1]__10viKVTE9W.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/EPD/Papers/ESA Papers/2025 [1]__10viKVTE9W.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2025 [2]__1LrJQqKVte.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/EPD/Papers/ESA Papers/2025 [2]__1LrJQqKVte.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare...
SRC00003	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare...
SRC00004	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0105. Sidecar variants compare...
SRC00006	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...
SRC00007	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...
SRC00009	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0066. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0067 to s0080. Sidecar variants compare...
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0073. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0074 to s0140. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0141 to s0145. Sidecar variants compare...
SRC00014	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...
SRC00015	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...
SRC00017	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0081. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0082 to s0164. Sidecar variants compare...
SRC00018	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare...
SRC00020	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare...
SRC00021	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0111. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...
SRC00023	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...
SRC00024	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...
SRC00026	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...
SRC00027	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...
SRC00029	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0063. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0064 to s0114. Sidecar variants compare...
SRC00030	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...
SRC00032	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...
SRC00033	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00034	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...
SRC00035	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...
SRC00037	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00038	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...
SRC00039	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...
SRC00040	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0069. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0070 to s0109. Sidecar variants compare...
SRC00042	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare...
SRC00044	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0130. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0131 to s0137. Sidecar variants compare...
SRC00046	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00048	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00047	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00049	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00051	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00050	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00052	0 covered · 0 excluded · 12 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00054	0 covered · 0 excluded · 12 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00053	0 covered · 0 excluded · 12 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00055	0 covered · 0 excluded · 16 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00057	0 covered · 0 excluded · 16 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00056	0 covered · 0 excluded · 16 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00058	0 covered · 0 excluded · 18 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00060	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00059	0 covered · 0 excluded · 18 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00061	0 covered · 0 excluded · 13 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00063	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00062	0 covered · 0 excluded · 13 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00064	0 covered · 0 excluded · 12 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00066	0 covered · 0 excluded · 12 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00065	0 covered · 0 excluded · 12 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00067	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00069	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00068	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00070	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00072	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00071	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00073	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00075	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00074	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00076	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00078	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00077	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00079	11 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00081	11 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00080	11 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00082	11 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00084	11 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00083	11 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00085	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
SRC00086	1 covered · 0 excluded · 0 gaps	q1: Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

C Revision and retrieval practice